4 Fig. 4.1 shows a mass spectrometer comprising a velocity selector (Stage 1) and an ion separator (Stage 2). A stream of alpha particles is sent into the mass spectrometer.

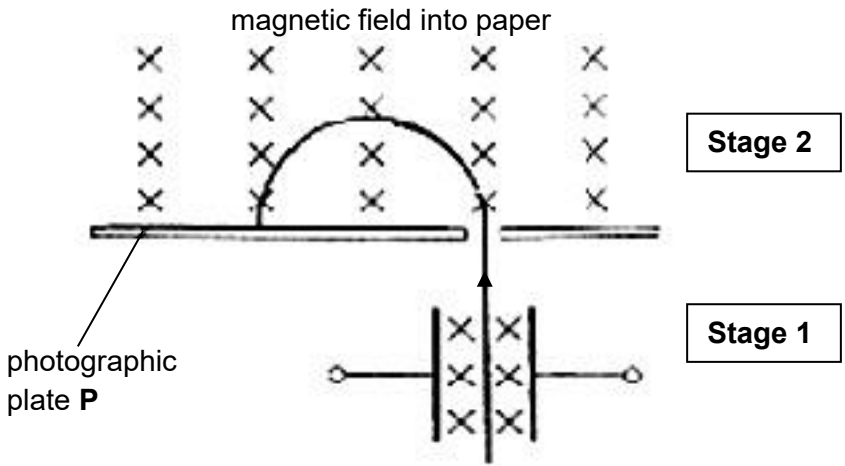
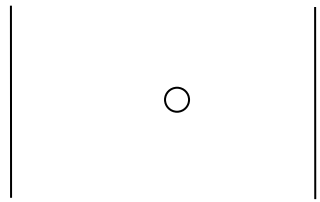


Fig. 4.1

(a) **Stage 1**

The alpha particles have a range of speeds enter Stage 1 which isolates particles with a specific speed, *v*. The plates producing the electric field have a separation of 1.0 cm. A uniform magnetic field is applied between the plates.

(a) **(i)** On the diagram below, draw arrows to represent the electric force F_E and the magnetic force F_B. [2]



(ii) Describe and explain the path of alpha particles with speed less than *v*.

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……[2]

(b) **Stage 2**

After Stage 1, alpha particles of speed 1.00×10^5 m s^{-1} are sent into stage 2. The alpha particles are travelling at right angles to the uniform magnetic field and are detected using a photographic plate P placed at the appropriate position. Determine the distance between the point of entry and the point of impact of the alpha particles with the photographic plate when the magnetic flux density 0.050 T is used.

distance = ……m [2]

(c) A similar spectrometer can be used with beta particle. Beta particles typically travel more than 100 times faster than the alpha particles. State and explain the necessary modifications to the spectrometer for it to work with beta particles.

Stage 1:

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Stage 2:

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[4]